CMSC6950 Homework #4

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Question 1

In [ ]:
```
# Import libraries
import pandas as pd
import glob
```

In [76]:
```
for filename in glob.glob("./data/gapminder*.csv"):

    print(filename)
    data = pd.read_csv(filename) # load data
    # print(data.columns)

    # Extract continent name from filename
    continent = filename.replace('.csv', '').split('_')[-1]

    gdp_cols = [col for col in data.columns if col.startswith('gdpPerc

    for col in gdp_cols:
        # print(col)
        year = col.split('_')[1] # get year

        q25, q75 = data[col].quantile([0.25, 0.75]) # compute quantile

        # Masks
        mask_wealthy = data[col] > q75
        mask_poor = data[col] < q25
        mask_middle = (data[col] >= q25) & (data[col] <= q75)

        mean_wealthy = data.loc[mask_wealthy, col].mean()
        mean_middle = data.loc[mask_middle, col].mean()
        mean_poor = data.loc[mask_poor, col].mean()

        print(f"{year} - {continent}")
        print('-'*30)
        print(f"Wealthy, mean={mean_wealthy:.2f}")
        print(f"Middle, mean={mean_middle:.2f}")
        print(f"Poor, mean={mean_poor:.2f}\n")
```

```
./data/gapminder_gdp_americas.csv
1952 - americas
------------------------------
Wealthy, mean=8377.01
```

```
Middle, mean=3082.78
Poor, mean=1939.73

1957 - americas
------------------------------
Wealthy, mean=9373.22
Middle, mean=3628.64
Poor, mean=1998.24

1962 - americas
------------------------------
Wealthy, mean=9340.21
Middle, mean=4148.16
Poor, mean=2095.21

1967 - americas
------------------------------
Wealthy, mean=11042.56
Middle, mean=4787.84
Poor, mean=2201.51

1972 - americas
------------------------------
Wealthy, mean=12880.31
Middle, mean=5373.77
Poor, mean=2523.75

1977 - americas
------------------------------
Wealthy, mean=14509.43
Middle, mean=6028.60
Poor, mean=3061.96

1982 - americas
------------------------------
Wealthy, mean=14687.07
Middle, mean=6217.45
Poor, mean=3119.87

1987 - americas
------------------------------
Wealthy, mean=16083.94
Middle, mean=6221.30
Poor, mean=2909.08

1992 - americas
------------------------------
Wealthy, mean=17083.86
Middle, mean=6285.37
Poor, mean=2818.41

1997 - americas
```

```
------------------------------
Wealthy, mean=18828.77
Middle, mean=7024.39
Poor, mean=2990.47

2002 - americas
------------------------------
Wealthy, mean=20710.58
Middle, mean=6871.00
Poor, mean=3100.90

2007 - americas
------------------------------
Wealthy, mean=23759.85
Middle, mean=8602.80
Poor, mean=3446.72

./data/gapminder_gdp_europe.csv
1952 - europe
------------------------------
Wealthy, mean=9697.72
Middle, mean=5234.45
Poor, mean=2370.97

1957 - europe
------------------------------
Wealthy, mean=11629.64
Middle, mean=6542.22
Poor, mean=3032.77

1962 - europe
------------------------------
Wealthy, mean=13619.49
Middle, mean=7996.97
Poor, mean=3756.38

1967 - europe
------------------------------
Wealthy, mean=16011.89
Middle, mean=9827.59
Poor, mean=4829.16

1972 - europe
------------------------------
Wealthy, mean=19125.39
Middle, mean=12310.27
Poor, mean=6130.05

1977 - europe
------------------------------
Wealthy, mean=21370.11
Middle, mean=14284.48
```

```
Poor, mean=7196.98

1982 - europe
------------------------------
Wealthy, mean=23207.74
Middle, mean=15858.85
Poor, mean=7606.38

1987 - europe
------------------------------
Wealthy, mean=26178.42
Middle, mean=17362.26
Poor, mean=7991.29

1992 - europe
------------------------------
Wealthy, mean=27912.77
Middle, mean=17266.52
Poor, mean=5851.70

1997 - europe
------------------------------
Wealthy, mean=30747.01
Middle, mean=19585.29
Poor, mean=6516.66

2002 - europe
------------------------------
Wealthy, mean=34149.99
Middle, mean=22858.36
Poor, mean=7266.88

2007 - europe
------------------------------
Wealthy, mean=38222.87
Middle, mean=26347.21
Poor, mean=9623.82

./data/gapminder_all.csv
1952 - all
------------------------------
Wealthy, mean=10041.52
Middle, mean=2098.03
Poor, mean=573.12

1957 - all
------------------------------
Wealthy, mean=11562.49
Middle, mean=2451.20
Poor, mean=630.07

1962 - all
```

```
------------------------------
Wealthy, mean=12418.34
Middle, mean=2846.91
Poor, mean=686.71

1967 - all
------------------------------
Wealthy, mean=14303.73
Middle, mean=3382.51
Poor, mean=749.15

1972 - all
------------------------------
Wealthy, mean=17848.34
Middle, mean=4141.45
Poor, mean=803.05

1977 - all
------------------------------
Wealthy, mean=18854.66
Middle, mean=4713.85
Poor, mean=825.89

1982 - all
------------------------------
Wealthy, mean=19156.65
Middle, mean=4970.40
Poor, mean=836.58

1987 - all
------------------------------
Wealthy, mean=20516.08
Middle, mean=5055.39
Poor, mean=818.74

1992 - all
------------------------------
Wealthy, mean=22051.60
Middle, mean=4792.50
Poor, mean=810.83

1997 - all
------------------------------
Wealthy, mean=24763.41
Middle, mean=5271.72
Poor, mean=841.71

2002 - all
------------------------------
Wealthy, mean=27302.46
Middle, mean=5629.61
Poor, mean=871.48
```

```
2007 - all
------------------------------
Wealthy, mean=31518.39
Middle, mean=6977.14
Poor, mean=986.34

./data/gapminder_gdp_oceania.csv
1952 - oceania
------------------------------
Wealthy, mean=10556.58
Middle, mean=nan
Poor, mean=10039.60

1957 - oceania
------------------------------
Wealthy, mean=12247.40
Middle, mean=nan
Poor, mean=10949.65

1962 - oceania
------------------------------
Wealthy, mean=13175.68
Middle, mean=nan
Poor, mean=12217.23

1967 - oceania
------------------------------
Wealthy, mean=14526.12
Middle, mean=nan
Poor, mean=14463.92

1972 - oceania
------------------------------
Wealthy, mean=16788.63
Middle, mean=nan
Poor, mean=16046.04

1977 - oceania
------------------------------
Wealthy, mean=18334.20
Middle, mean=nan
Poor, mean=16233.72

1982 - oceania
------------------------------
Wealthy, mean=19477.01
Middle, mean=nan
Poor, mean=17632.41

1987 - oceania
------------------------------
```

```
Wealthy, mean=21888.89
Middle, mean=nan
Poor, mean=19007.19

1992 - oceania
------------------------------
Wealthy, mean=23424.77
Middle, mean=nan
Poor, mean=18363.32

1997 - oceania
------------------------------
Wealthy, mean=26997.94
Middle, mean=nan
Poor, mean=21050.41

2002 - oceania
------------------------------
Wealthy, mean=30687.75
Middle, mean=nan
Poor, mean=23189.80

2007 - oceania
------------------------------
Wealthy, mean=34435.37
Middle, mean=nan
Poor, mean=25185.01

./data/gapminder_gdp_africa.csv
1952 - africa
------------------------------
Wealthy, mean=2619.57
Middle, mean=995.73
Poor, mean=399.26

1957 - africa
------------------------------
Wealthy, mean=2950.77
Middle, mean=1069.89
Poor, mean=450.39

1962 - africa
------------------------------
Wealthy, mean=3529.52
Middle, mean=1180.95
Poor, mean=500.90

1967 - africa
------------------------------
Wealthy, mean=5046.27
Middle, mean=1290.78
Poor, mean=573.63
```

```
1972 - africa
------------------------------
Wealthy, mean=5892.94
Middle, mean=1413.71
Poor, mean=638.10

1977 - africa
------------------------------
Wealthy, mean=6769.99
Middle, mean=1459.35
Poor, mean=655.06

1982 - africa
------------------------------
Wealthy, mean=6400.53
Middle, mean=1436.20
Poor, mean=653.45

1987 - africa
------------------------------
Wealthy, mean=5837.75
Middle, mean=1340.87
Poor, mean=611.18

1992 - africa
------------------------------
Wealthy, mean=5997.96
Middle, mean=1265.99
Poor, mean=597.29

1997 - africa
------------------------------
Wealthy, mean=6300.32
Middle, mean=1307.99
Poor, mean=598.74

2002 - africa
------------------------------
Wealthy, mean=7020.19
Middle, mean=1390.17
Poor, mean=597.01

2007 - africa
------------------------------
Wealthy, mean=8531.07
Middle, mean=1603.79
Poor, mean=617.48

./data/gapminder_gdp_asia.csv
1952 - asia
------------------------------
```

```
Wealthy, mean=18003.93
Middle, mean=1367.84
Poor, mean=520.78

1957 - asia
------------------------------
Wealthy, mean=19870.96
Middle, mean=1611.44
Poor, mean=579.13

1962 - asia
------------------------------
Wealthy, mean=19033.63
Middle, mean=1873.83
Poor, mean=618.13

1967 - asia
------------------------------
Wealthy, mean=18997.57
Middle, mean=2357.95
Poor, mean=622.89

1972 - asia
------------------------------
Wealthy, mean=26228.45
Middle, mean=3260.87
Poor, mean=615.51

1977 - asia
------------------------------
Wealthy, mean=22639.44
Middle, mean=4158.45
Poor, mean=663.02

1982 - asia
------------------------------
Wealthy, mean=20407.14
Middle, mean=4477.77
Poor, mean=743.40

1987 - asia
------------------------------
Wealthy, mean=20544.30
Middle, mean=4711.29
Poor, mean=828.14

1992 - asia
------------------------------
Wealthy, mean=23978.84
Middle, mean=5062.10
Poor, mean=902.93
```

```
1997 - asia
------------------------------
Wealthy, mean=26624.56
Middle, mean=6072.05
Poor, mean=1037.97

2002 - asia
------------------------------
Wealthy, mean=27283.28
Middle, mean=6346.66
Poor, mean=1198.19

2007 - asia
------------------------------
Wealthy, mean=34152.12
Middle, mean=7409.56
Poor, mean=1553.80
```

Question 2

Preamble:

We use **FOR** loops when iterating

- over a list (or tuple)
- over known number of iterations

We use **WHILE** loops when iterating

- over an unknown number of iterations
- until a condition is met

(a) Determining how many numbers in a given list are odd

- For a "given list" it would be easier to go with a `for` loop.
- A `for` loop automatically iterates through each element in the list.
- On the other hand, in a `while` loop we would have to check the length of list and loop over the index of the elements. This method is more error-prone.

Since we know exactly what to iterate over `for` is a better choice.

(b) Finding the fifth odd number in a given list (if there are at least five such numbers in the list)

- We can iterate through the list, count odd numbers, and stop when you reach the fifth one:
- A `while` loop could work, similar the (a) above, it quickly becomes complex

and error prone to implement.

(c) Finding the first time the product of two consecutive numbers in a list is greater than 20 and at least one number is positive (if this occurs in the list)

- We could use both methods but a `for` loop is the better choice because we are iterating through a known collection.
- A `while` loop would work but adds unnecessary complexity (long code).

In [113…

```
list = [12, 3, 2, 4, 6, 9, 11, 5, 7, 3]

count = 0
for n in list:
    print(f"Iterate element: {n}")
    if n % 2 == 1:
        print(f"    {n} is an odd number")
        count += 1
        print(f"    Adding to count")
    print(f"Total odd number count: {count}\n")

print(f"Number of odd number in {list} is {count}")
```

```
Iterate element: 12
Total odd number count: 0

Iterate element: 3
    3 is an odd number
    Adding to count
Total odd number count: 1

Iterate element: 2
Total odd number count: 1

Iterate element: 4
Total odd number count: 1

Iterate element: 6
Total odd number count: 1

Iterate element: 9
    9 is an odd number
    Adding to count
Total odd number count: 2

Iterate element: 11
    11 is an odd number
    Adding to count
Total odd number count: 3

Iterate element: 5
    5 is an odd number
    Adding to count
Total odd number count: 4

Iterate element: 7
    7 is an odd number
    Adding to count
Total odd number count: 5

Iterate element: 3
    3 is an odd number
    Adding to count
Total odd number count: 6

Number of odd number in [12, 3, 2, 4, 6, 9, 11, 5, 7, 3] is 6
```

In [ ]: